AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 20 · STUDENT LESSON

Privacy and data governance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Name the purpose. Limit the fields. Trace the readers.

Audit fictional governance plans without confusing codes, metadata checks and privacy guarantees.

Keep one habit: Purpose contract → field ledger → native key-separation diagram → control-evidence probes → plan audit → independent subgroup rule.

MAIN LESSON · SECTION 1

Follow purpose, fields, readers and retention through a data plan

> Removing names is not the same as establishing anonymity.

Learning goals

I can explain the **mechanism** and terminology of privacy and **data governance**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve permission, source identity and evidence boundaries

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting response

- Recall Research ethics and consent. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

My starting explanation

End goal

Explain privacy and **data governance** using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of privacy and **data governance**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Audit the complete data flow rather than the presence of names

> Separate purpose limitation, field minimization and verified protection.

Begin with an identifying combination

Imagine an export without names that still contains exact birth date, classroom and a rare condition. A reader who knows those details may recognize someone. Governance asks why each field is needed, who can read each table, how records can be linked and when they should be removed. In this fictional exercise, audit the declared plan rather than handling personal records.

A clear plan is not evidence that technical restrictions or deletion have been implemented.

Original governance explanation

Data governance specifies who may use which records, for what purpose and for how long. Coded data can still be reidentifiable.

Full declared-purpose and access contract

Governance item	Complete fictional exercise contract
Named purpose	Task-score analysis using fictional records
Analysis fields needed in the picture	Random participant code and task score
Unnecessary main-export details	Exact birth date, classroom and rare condition are not needed for this toy purpose
Separate key	Code-to-identity link; restricted withdrawal role only
Analyst role	Reads analysis values, not the identity key under this design
Public output in the proposed repair	No row-level public output; any later aggregate needs its own disclosure check

Governance item	Complete fictional exercise contract
Retention in the proposed repair	Demo plan: delete analysis and key at exercise end; this is a proposal, not executed deletion
Subgroup policy for the independent case	Do not report groups smaller than five
Evidence limit	Metadata checks do not prove anonymity, actual access control, consent or safe export

Supplied and missing evidence

> **List the given facts and one fact that is not supplied.**

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace how an unnamed record can remain linkable

> A field-by-field audit precedes any proposed export.

Original minimization and separation procedure

For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.

Complete main-field audit

Main export field	Reason for concern	Toy-purpose action
Coded ID	May link to other records or a key	Restrict under the declared role plan
Exact birth date	Detailed identifying combination	Remove from the task-score analysis proposal
Classroom	Narrows the possible people	Remove unnecessary detail
Rare condition	Can identify through unusual combinations	Remove from this toy-purpose proposal
No names in export	Does not remove all linkage routes	Do not label it guaranteed anonymous
Missing readers/retention evidence	Plan alone does not prove controls	Record unresolved checks; do not claim actual protection

Preserve scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Privacy and data governance — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed.	For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.	The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict publishing the identity key with the analysis

> The change breaks the stated separation design.

Main task and response

- Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the changed key access

- Change: Publish the identity key with the analysis table.

▼ Check after predicting

Prediction: The records become directly linkable to names and breach this access design.

Reason: Separation only helps while the linkage key remains restricted.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the linking key, allowed readers and retention together

> A separate file is useful only when its access design is respected.

Main coded-export claim and checked result

Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed.

The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate code/key facts and limits

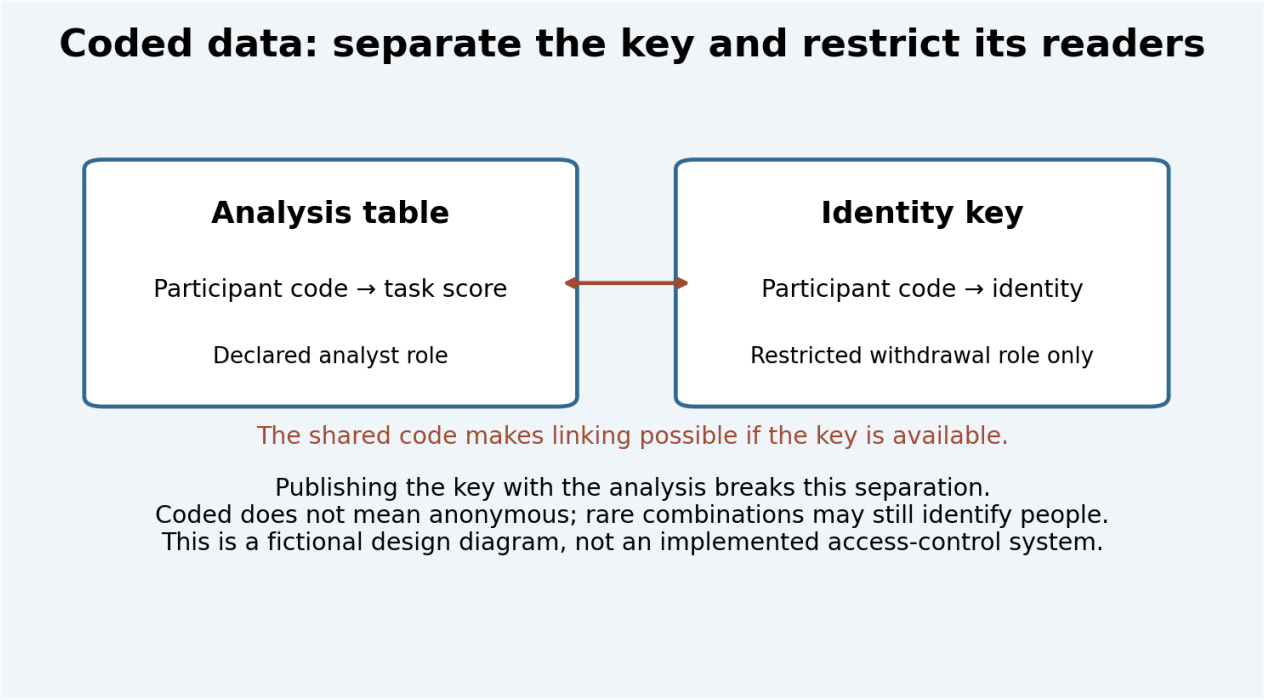
Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.

The analysis table holds codes and scores. A separate restricted key links codes to names only for authorised withdrawal handling.

The separate picture keeps participant codes and task scores in the analysis table. A restricted key maps those codes to identities for the stated withdrawal purpose. That link is why the records are pseudonymous rather than guaranteed anonymous. The separation reduces routine exposure only if readers actually have the declared restrictions.

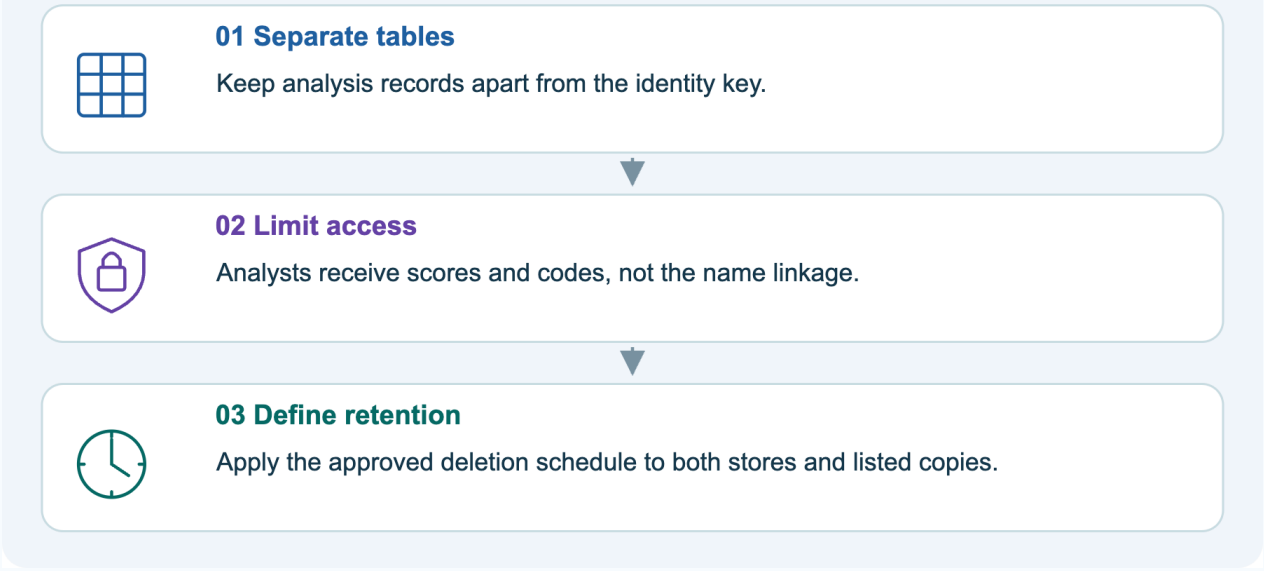
Publishing the key alongside the analysis makes code-based linking straightforward and breaks this design. Even with a restricted key, unusual values or small groups may disclose identity through other information. No actual identity lookup, access-control enforcement or deletion is performed here.

See the native linking diagram



AI ACADEMY / AI-7/20 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Separate tables	Keep analysis records apart from the identity key.
2. Limit access	Analysts receive scores and codes, not the name linkage.
3. Define retention	Apply the approved deletion schedule to both stores and listed copies.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Pseudonymous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

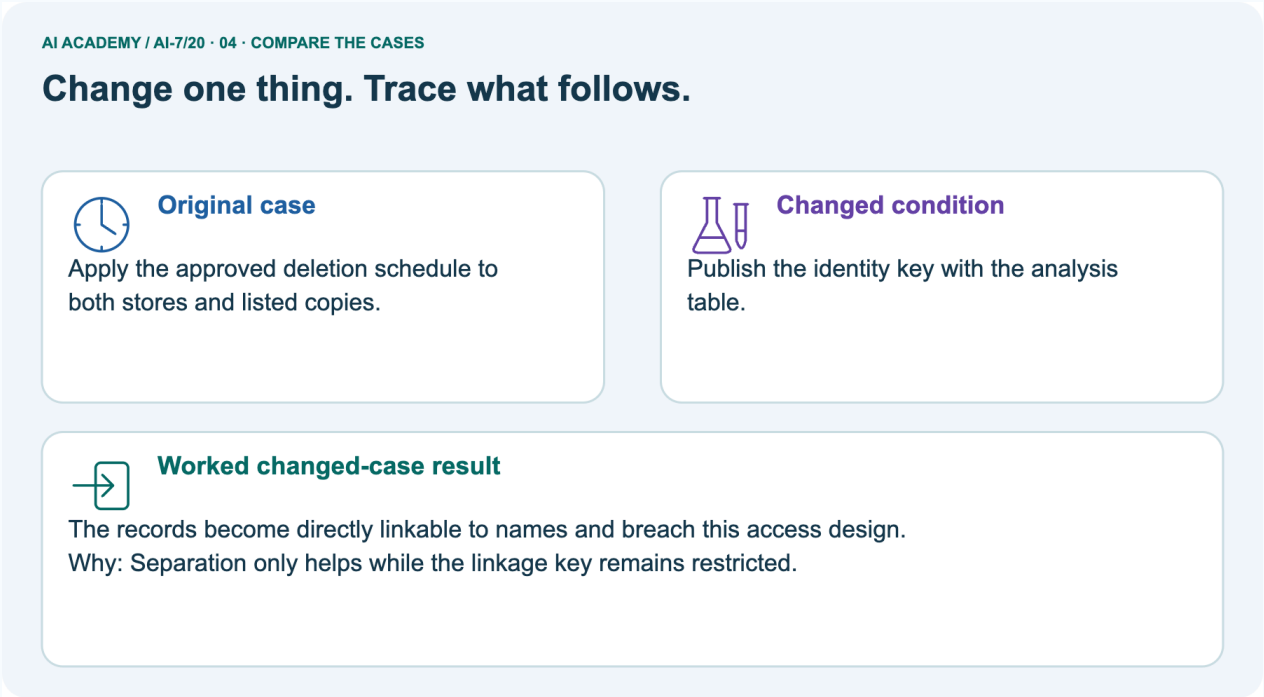
Complete role and protection-evidence audit

Table or operation	Declared permitted role/use	What the audit cannot assume
Analysis code + score	Analyst for the named task-score purpose	Codes are not proof of anonymity
Identity key	Restricted withdrawal role	A table label does not enforce permissions
Combine key and analysis publicly	Outside this design	Do not treat the result as deidentified
End-of-exercise deletion plan	Proposed retention action	Not evidence deletion has occurred
Subgroup count below five	Blocked by supplied policy	Changing the threshold after seeing outcomes is not compliance
Subgroup count equals five	Meets this size rule only	Not a general privacy or export guarantee

Predict publishing the key

- Change one thing: Publish the identity key with the analysis table. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain remaining linkage risks

- Small groups or unusual records may still be identifiable without names; evaluate re-identification risk.

- Read the labelled picture: Point to the analysis table and the restricted identity key. Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

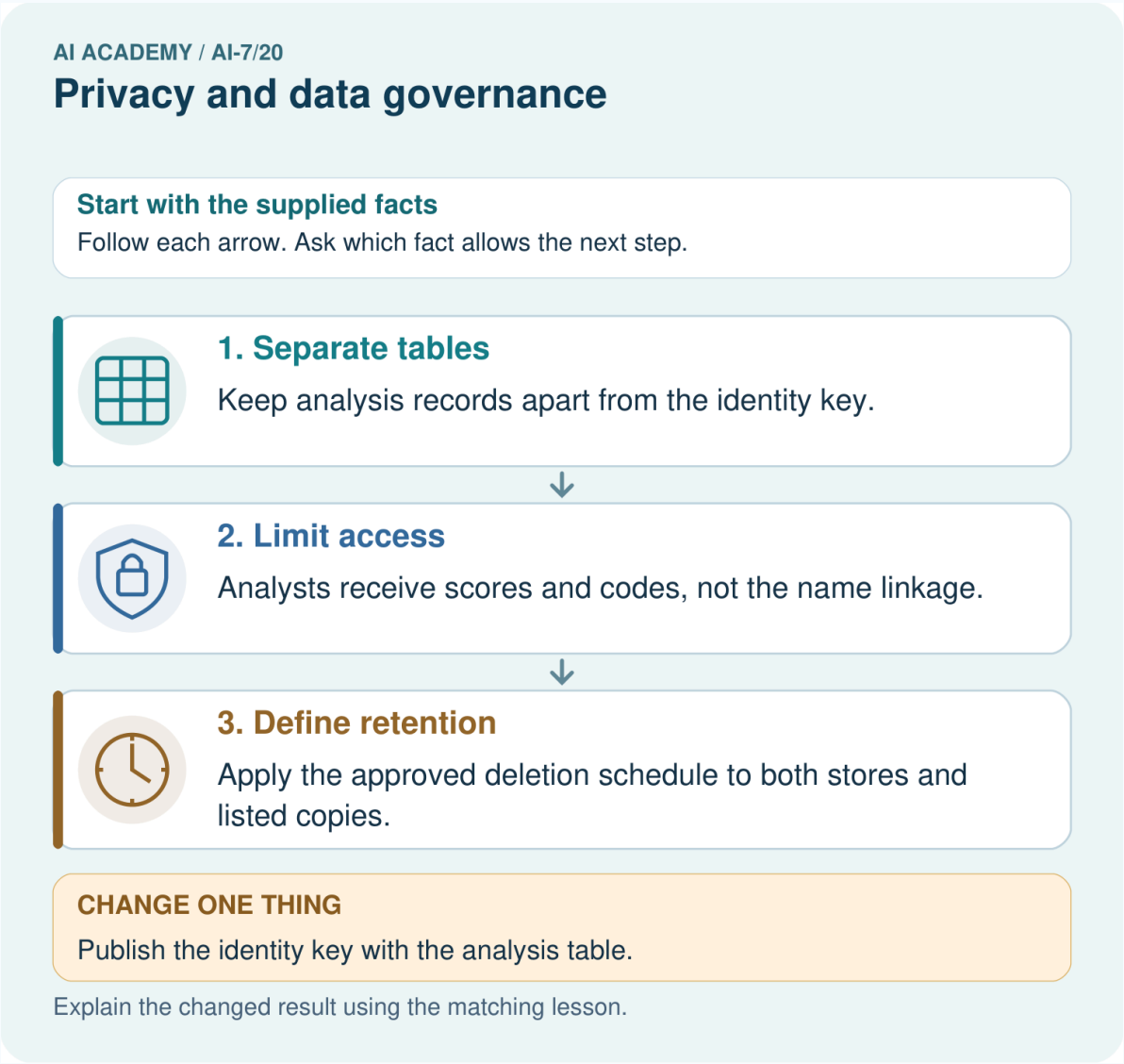
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Privacy and data governance

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair anonymity claims and single-rule privacy shortcuts

> A proposed control needs implementation evidence and context.

Original qualification and repair setup

A size threshold or coding scheme alone does not guarantee privacy.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete access, retention and size-rule probes

Governance probe	Checked interpretation
Names removed	Other fields may still identify people
Random codes used	Key or external information can allow linkage
Key is in another file	File separation does not prove reader restrictions
Metadata audit returns no listed issues	Not actual permission, anonymity or disclosure clearance
Retention action written down	Not proof of deletion
One-member subgroup under minimum-five policy	Do not publish the subgroup
Five-member subgroup	Meets only the stated size rule; other risks remain
Merge or suppress categories	Check permitted purpose and whether totals reveal the suppressed cell
Fictional example is well designed	Does not authorize a real export

Write a justified repair

- A report omits this qualification: A size threshold or coding scheme alone does not guarantee privacy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise checking a declared plan without using personal records

> Reserve the one-person subgroup scenario for the first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported workbook trace

Use workbook W2 for the connected example "Privacy and **data governance**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual check

- Use the worked case for Privacy and **data governance**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Point to the analysis table and the restricted identity key. Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

My explanation using a pictured step and the given facts

Reserve selected subgroup case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a declared-plan audit with explicit limits

> Check metadata without pretending a real protection system exists.

Current fictional proposal

Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed. Show the intermediate steps and give the conclusion with its scope.

Complete executable metadata checks

```
"""Audit a declared fictional data-flow plan, not actual privacy or permission."""

def audit_plan(plan):
 required={'analysis_fields','analyst_can_read_key','public_fields','purpose','retention'}
 if not isinstance(plan,dict) or not required<=plan.keys():raise ValueError('complete declared plan
required')
 if type(plan['analyst_can_read_key']) is not bool:raise ValueError('key access must be Boolean')
```

```
 for key in ['analysis_fields','public_fields']:
  if not isinstance(plan[key],list) or any(not isinstance(v,str) for v in plan[key]):raise
ValueError('field lists required')
 issues=[]
 unnecessary=set(plan['analysis_fields'])-{'participant_code','task_score'}
 if unnecessary:issues.append('analysis includes unnecessary fields under this toy purpose')
 if plan['analyst_can_read_key']:issues.append('analyst access breaks the declared key separation')
 if set(plan['public_fields']) &
{'name','identity_key','participant_code','exact_birth_date','classroom','rare_condition','task_score'}:
  issues.append('public proposal contains row-level or linking fields outside the toy aggregate
plan')
 if not isinstance(plan['purpose'],str) or not plan['purpose'].strip():issues.append('purpose
unspecified')
 if not isinstance(plan['retention'],str) or not plan['retention'].strip():issues.append('retention
unspecified')
 return issues
main={'analysis_fields':['participant_code','exact_birth_date','classroom','rare_condition'],
      'analyst_can_read_key':False,'public_fields':
['participant_code','exact_birth_date','classroom','rare_condition'],
      'purpose':'fictional task-score analysis','retention':''}
# Explicit proposed repair; passing these metadata checks is not authorization.
proposed={'analysis_fields':['participant_code','task_score'],'analyst_can_read_key':False,
          'public_fields':[],'purpose':'fictional task-score analysis',
          'retention':'demo policy: delete analysis and key at end of the named exercise; not an
executed deletion'}
main_issues=audit_plan(main);proposed_issues=audit_plan(proposed)
published_key={**proposed,'public_fields':['identity_key','participant_code','task_score']}
assert main_issues and not proposed_issues and audit_plan(published_key)
def subgroup_rule(size,minimum=5):
 if type(size) is not int or type(minimum) is not int or size<0 or minimum<1:raise
ValueError('nonnegative count and positive minimum required')
 return 'block_under_supplied_rule' if size<minimum else 'size_rule_met_only'
assert subgroup_rule(1)=='block_under_supplied_rule' and subgroup_rule(5)=='size_rule_met_only'
print('Main proposal issues:',main_issues)
print('Proposed repair metadata issues:',proposed_issues,'does not mean approved or anonymous')
print('Published-key proposal issues:',audit_plan(published_key))
print('Supplied minimum-five rule: one-person subgroup blocked; five meets only this count rule, not
a privacy guarantee.')
print('No personal records, identity lookup, export, access control or deletion is executed. The
audit checks declared metadata, not actual reader permissions, consent or reidentification risk.')
```

Expected issues and scope

1. Predict which declared fields, readers and missing retention details the main proposal violates. The full code audits plan metadata, not actual people or permissions. A proposed repair keeps code and score analysis, separates the key role, includes no public row-level output and states an example retention action. No listed issues means only those checks passed. It does not establish authorization or anonymity.

   A separate size-rule check includes equality at five, without treating that boundary as a privacy guarantee. No export, identity lookup or deletion occurs.

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

Try alone: apply the minimum-five subgroup policy

> Decide the supplied case without claiming the rule guarantees privacy.

Independent subgroup case

Aggregate contains one member in a subgroup; policy forbids subgroup reports smaller than 5. Decide.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Aggregate contains one member in a subgroup; policy forbids subgroup reports smaller than 5. Decide.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save purpose, roles, field choices and retention evidence

> Keep proposed controls separate from implemented controls.

Project and artifact

Add a labelled privacy and **data governance evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of privacy and **data governance**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local metadata audit or paper trace

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain why a coded record can remain identifiable

> State the purpose, readers, key access and remaining risks.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of privacy and **data governance**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Data governance	Rules for purpose, access, quality and lifetime
Pseudonym	A code replacing direct identification
Disclosure risk	The possibility of learning identifying or sensitive facts

Use two terms to explain a step in this week's example.

▼ **Extra challenge · Explain a stronger case**

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.
- A report omits this qualification: A size threshold or coding scheme alone does not guarantee privacy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.
- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ **Reference · Check the research reasoning and its limits**

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Privacy and data governance — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed.	For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.	The combination can remain identifying; coded is not necessarily anonymous. Minimise fields and assess disclosure under the declared policy.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.

By the end, I can	How I will show it
Explain the mechanism and terminology of privacy and data governance	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.

- Which labelled step matches this detail? Apply the approved deletion schedule to both stores and listed copies.
- 1. Define retention
- 2. Separate tables
- 3. Limit access
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Define retention Apply the approved deletion schedule to both stores and listed copies.

Pseudonymous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

Changed-case question and optional hint

- Changed-case practice: Publish the identity key with the analysis table. Which conclusion is supported by the supplied facts?
- 1. The records become directly linkable to names and breach this access design.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The records become directly linkable to names and breach this access design. Separation only helps while the linkage key remains restricted.

Repair and retain the record

Review Research ethics and consent (ai-7/19). Goal: Explain the **mechanism** and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Privacy and data governance: the case



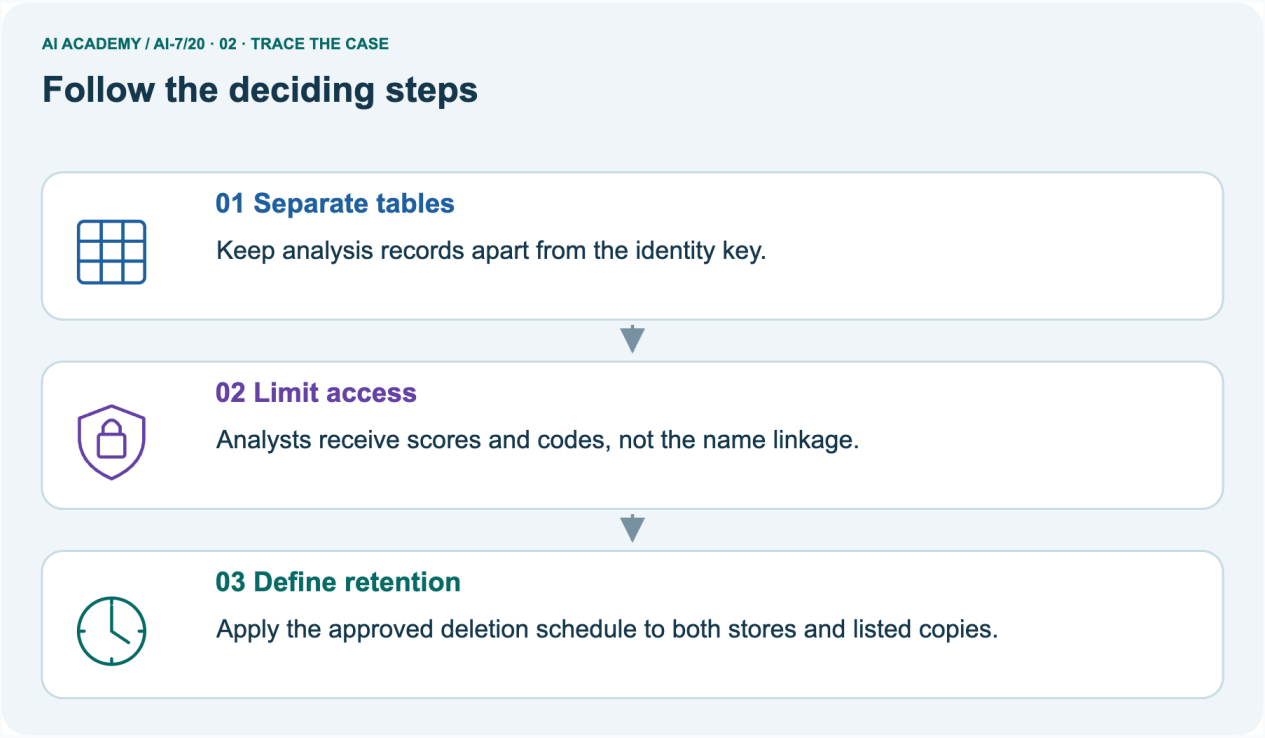
Follow the labelled picture with the matching explanation. Open the vector diagram

Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names. Supplied facts: The analysis table holds codes and scores.

A separate restricted key links codes to names only for authorised withdrawal handling.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

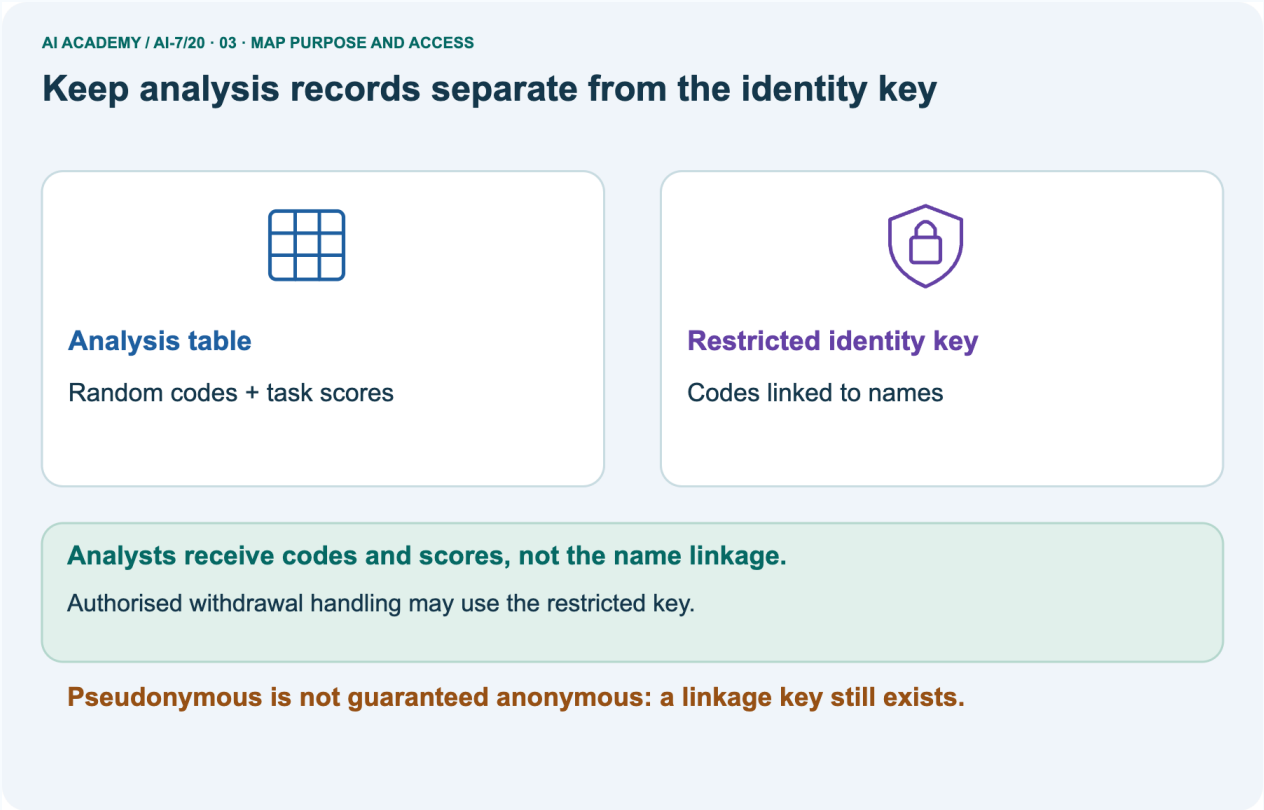


Follow the labelled picture with the matching explanation. Open the vector diagram

Separate tables: Keep analysis records apart from the identity key. → Limit access: Analysts receive scores and codes, not the name linkage. → Define retention: Apply the approved deletion schedule to both stores and listed copies.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep analysis records separate from the identity key

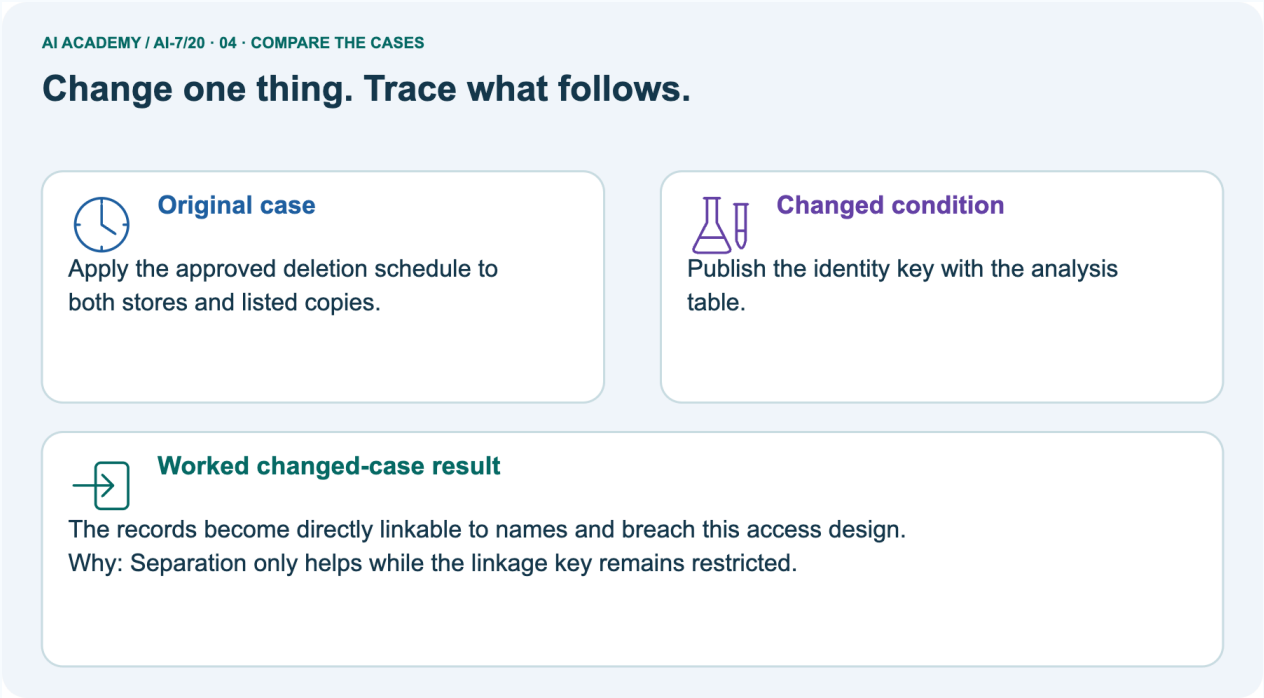


Follow the labelled picture with the matching explanation. Open the vector diagram

Analysts receive random participant codes and scores. A separate restricted key links codes to names only for authorised withdrawal handling. Keeping the key restricted reduces exposure; it does not guarantee anonymity.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



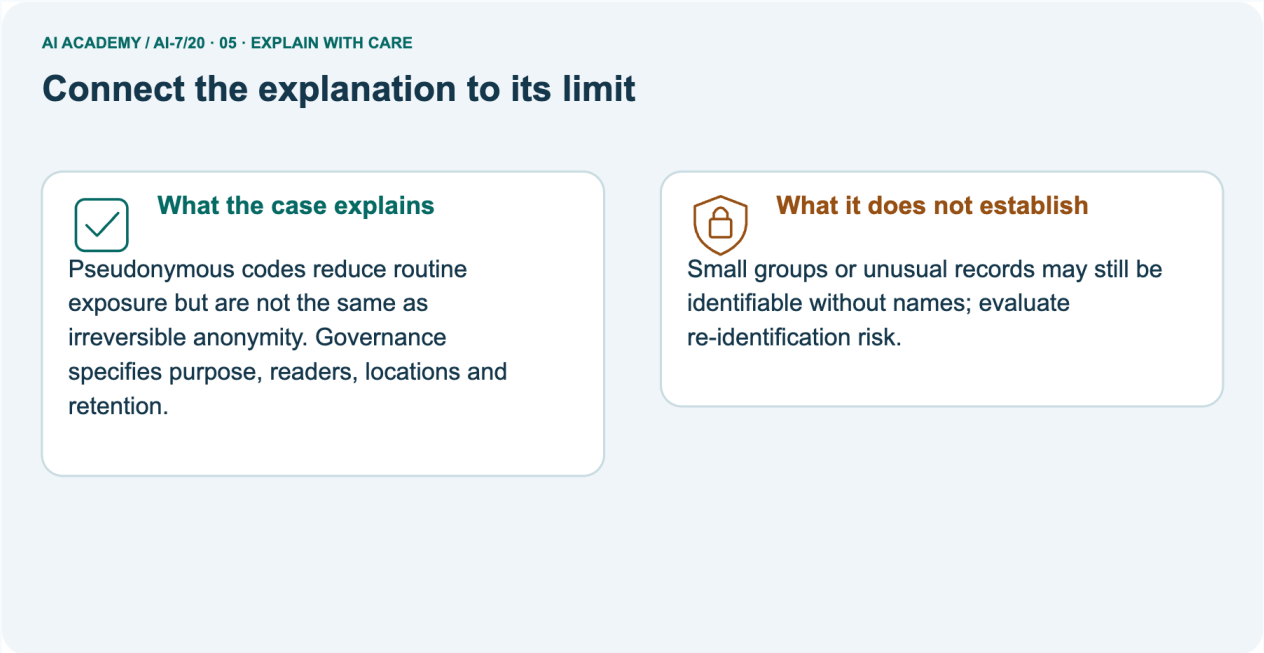
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Apply the approved deletion schedule to both stores and listed copies.
Changed condition: Publish the identity key with the analysis table.

Expected result: The records become directly linkable to names and breach this access design. Reason: Separation only helps while the linkage key remains restricted.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



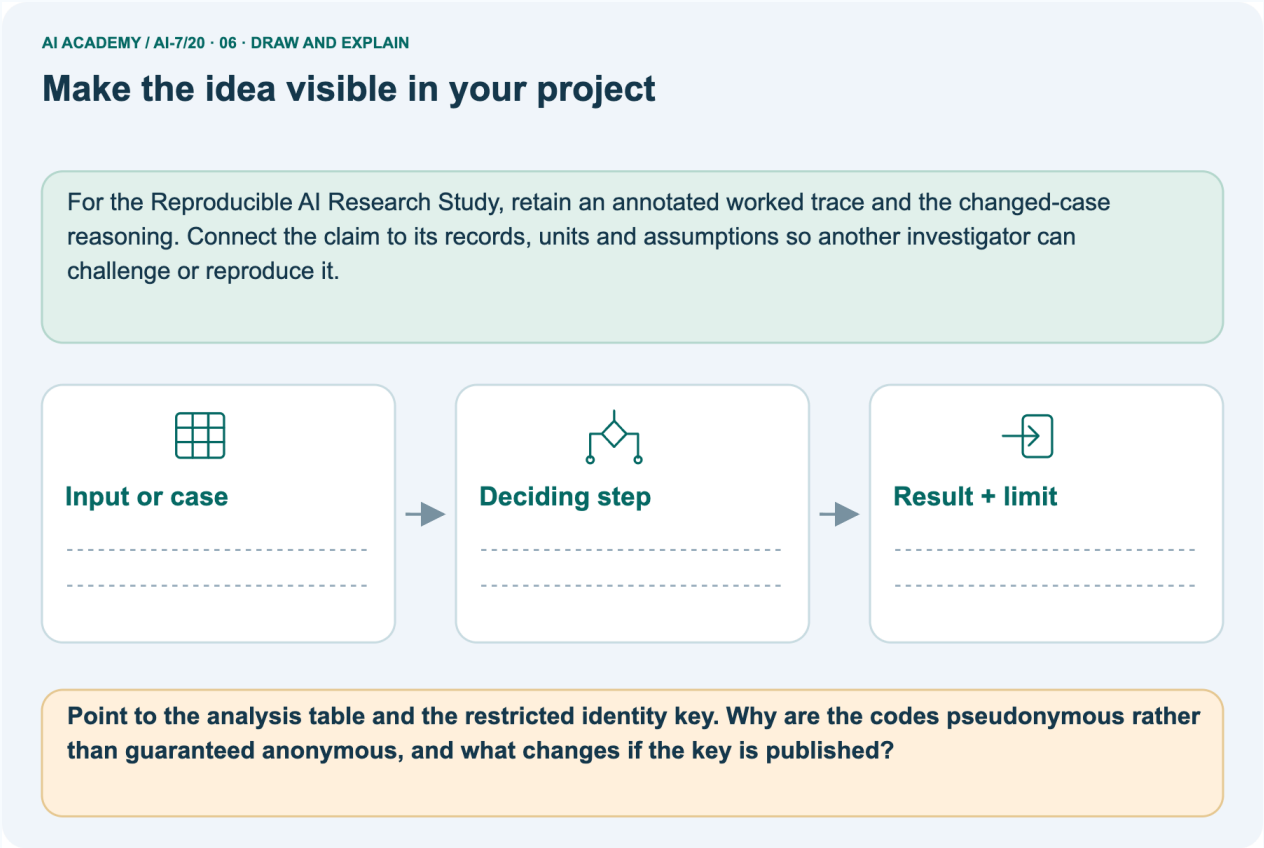
Follow the labelled picture with the matching explanation. Open the vector diagram

Pseudonymous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

Limit: Small groups or unusual records may still be identifiable without names; evaluate re-identification risk.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the analysis table and the restricted identity key.

Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Privacy and data governance: the case

02. Follow the deciding steps

03. Keep analysis records separate from the identity key

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.